Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

Cortical representations *drift*: a decoder fit to one animal's neural activity slowly loses accuracy on that same animal over days, even when behavior is unchanged. The standard remedy is periodic recalibration on fresh within-subject data. We report a different form of robustness. On the open BraiDyn-BC cohort (25 mice, cortex-wide widefield $\Delta F/F$ parcellated into 44 Allen-atlas regions, a $\sim$15-day operant protocol), we decode four task events and compare, per mouse, how much accuracy is lost across two weeks by (i) that mouse's *own* early-week decoder and (ii) a decoder *pooled* over the other 24 mice's early-week data. The pooled decoder ages substantially less: the paired personal-minus-pooled aging asymmetry is $+0.024$ AUC ($p = 2 \times 10^{-5}$, sign-flip; the own decoder ages more in 76% of the 100 mouse–target pairs), and the effect holds for all four events individually ($p < 0.005$). A count-matched control is decisive about *why*: holding the number of training events fixed, a decoder built from *other* individuals ages far less than the subject's own ($+0.023$ AUC for self vs. -0.006 for a single other mouse at equal sample count), so the robustness is not a data-volume artifact but follows from not overfitting the subject's own drift-prone idiosyncrasies. Pooling over *many* other individuals rather than one adds a further, smaller and direction-consistent gain (to -0.012), yielding a graded self $\rightarrow$ one-other $\rightarrow$ many-others ordering. Population pooling is thus a drift-mitigation strategy in its own right: the small early-week accuracy cost of decoding across individuals is repaid as temporal stability. Code and a streamed-feature pipeline are released.

1 Introduction

Representational drift—the gradual reorganization of a neural code over days while the animal's behavior stays fixed—is now documented across sensory, motor, and association cortex and in hippocampus [14, 2, 16]. Its practical consequence for neural decoding is stark: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is *recalibration*—periodically refitting or adapting the decoder on fresh within-subject data [1, 4]—which requires ongoing labelled data from the individual and treats each subject in isolation.

We ask whether the code's cross-individual structure offers a complementary route to temporal robustness. Cortex-wide behavioral representations are known to be substantially shared across individuals [9, 6, 7], and recent models decode behavior zero-shot on unseen subjects [17]. That prior work is evaluated at essentially one point in time. The question we pose is temporal: *does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself?* If individual brains drift along idiosyncratic, partly independent trajectories, then averaging over many individuals should cancel the individual-specific component of drift—so a pooled de-

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coder would age more slowly than a personal one. This is a concrete, testable prediction that, to our knowledge, has not been examined.

We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose $\sim$15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift.** Across two weeks, a population-pooled decoder loses less accuracy than a mouse's own decoder, for all four decoded events (paired asymmetry $+0.024$ AUC, $p = 2 \times 10^{-5}$; Section 4.1).
- **It is not a data-volume artifact.** A count-matched control shows that, at equal training-set size, other-individual data ages far less than the subject's own; a graded self $\rightarrow$ one-other $\rightarrow$ many-others ordering isolates avoiding self-overfitting as the primary driver and cross-individual diversity as a secondary, consistent refinement (Sections 4.2–4.3).
- **A drift-mitigation framing.** The early-week accuracy penalty of cross-subject decoding is repaid as late-week stability, suggesting population priors as an alternative to per-subject recalibration. We release code and a no-raw-download streaming pipeline.

2 Related work

Representational drift and its mitigation. Drift is characterized within a single subject [14, 2, 16, 13], and the engineering response is within-subject recalibration or latent re-alignment: manifold stabilization [1], supervised and self-calibrating recalibration [4, 12], adversarial and nonlinear latent-dynamics alignment to the subject's own reference day [3, 11]. Every one of these stabilizes a decoder using *that subject's* own later data; none uses a population of *other* subjects as the source of stability, which is our question.

Cross-subject decoding. Shared structure across individuals supports cross-subject readout via manifold alignment [8], contrastive embeddings [15], and, on this very cohort, atlas-aligned foundation models [17]. Pooling across participants is also known to *reduce overfitting*: HTNet trains one decoder on pooled multi-participant data precisely to "avoid overfit decoders" from single-subject data scarcity [10]. Two things separate that line from ours. First, the overfitting it avoids is *sample-size* overfitting (too little data per subject), whereas our count-matched control holds training-set size fixed and still finds the effect, so the overfitting we implicate is to the subject's *drift-prone* features rather than to small samples. Second, and more basically, all of these target accuracy/transfer at a *fixed time*; the temporal-robustness question—whether the pooled decoder *ages* more slowly over days—is orthogonal and, to our knowledge, unaddressed.

Conservation of cortex-wide dynamics. That cortex-wide activity is dominated by shared, low-dimensional, movement-related components [9, 6, 7] is the premise that makes cross-subject pooling viable at all; it does not by itself imply anything about drift.

3 Data and methods

Dataset. BraiDyn-BC [5] (DANDI:001425, CC-BY): per mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions at $\sim$30 Hz, with time-aligned behavioral channels. Twenty-five mice each performed $\sim$15 daily operant sessions over $\sim$3 weeks. We stream only the 44-region traces and behavioral channels (never the raw movies).

Features and events. For each of four events (lever-pull, lick, reward, tone) we detect onsets as rising edges (1 s refractory) and form a 44-dimensional evoked feature: mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (-0.5–0 s) baseline. Negatives are matched-count windows drawn > 2 s from any event. The decoder is a standardized ℓ_2-regularized logistic regression throughout.

Temporal blocks. Each mouse's task sessions are pooled into an *early* block (days 1–5) and a *late* block (days 11–15). We report AUC in four regimes: within-mouse same-block (5-fold CV on early, WS), within-mouse across-block (train early $\rightarrow$ test late, WX), pooled same-block (train other mice's early $\rightarrow$ test held-out early, LS), and pooled across-block (train other mice's early $\rightarrow$ test held-out *late*, LX).

Table 1: Personal vs. pooled aging across days (AUC lost from early- to late-block test, holding training fixed), per event, $n = 25$ mice. Asymmetry = personal − pooled aging; positive ⇒ the mouse's own decoder ages more. p: paired sign-flip permutation.

Event	Personal aging	Pooled aging	Asymmetry (95% CI)	p	own ages more
Lever-pull	-0.012	-0.026	$+0.014$	0.002	68%
Lick	$+0.018$	-0.010	$+0.028$	0.0001	84%
Reward	$+0.023$	-0.010	$+0.033$	0.0001	80%
Tone	$+0.013$	-0.010	$+0.023$	0.005	72%
Pooled			$\mathbf{+0.024}$	$\mathbf{2 \times 10^{-5}}$	**76%**

Aging. We define a decoder's *aging* as its accuracy drop when the test set moves from the held-out mouse's early block to its late block, holding training fixed. Personal aging is $\mathrm{WS} - \mathrm{WX}$; pooled aging is $\mathrm{LS} - \mathrm{LX}$. The per-mouse *asymmetry* is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; positive means the own decoder ages more. Significance is a paired sign-flip permutation test over mice (20,000 permutations), with a cluster (over-mice) bootstrap 95% CI.

4 Results

4.1 A population-pooled decoder ages more slowly than a personal one

For every mouse we hold the test data fixed—that mouse's own early then late block—and vary only the decoder's *training* source: the mouse itself versus the other 24. The pooled decoder loses less accuracy across the two weeks (Table 1). The paired personal-minus-pooled aging asymmetry is positive for all four events (lever-pull $+0.014$, lick $+0.028$, reward $+0.033$, tone $+0.023$), each significant at $p < 0.005$; pooled over all 100 mouse–target pairs the asymmetry is $+0.024$ AUC ($p = 2 \times 10^{-5}$), and the personal decoder ages more in 76% of pairs. The sign is consistent even where both decoders happen to *improve* with practice (lever-pull): the pooled decoder improves *more*. Read as a gap, the early-week advantage of the personal decoder over the pooled one shrinks toward zero by the late week, because only the personal decoder ages.

4.2 The robustness is not a data-volume artifact

A pooled decoder is trained on $\sim 24\times$ more events than a personal one, so the obvious deflationary explanation is that more data, not pooling, buys the robustness. We rule this out with a count-matched control (Fig. 1, left). For each held-out mouse we fix the number of training events at that mouse's *own* early-block count n and compare three decoders on the same held-out early→late aging: **self** (the mouse's own n events), **one** (n events from a single other mouse), and **many** (n events drawn across the other 24). The result is a clean ordering, consistent in all four events (means over targets): self ages $+0.023$, one-other -0.006, many-others -0.012 AUC. Two things follow. First, at *identical* sample count the subject's own decoder ages $+0.029$ more than a single other mouse's—so volume is not the driver; training on data other than the subject's own is. The subject's early-week decoder overfits idiosyncratic features that then drift, and any decoder that never saw the subject sidesteps this. Second, pooling over many others rather than one ages a further -0.006 (many $<$ one in $4/4$ events), so cross-individual diversity contributes a smaller, secondary refinement on top of the primary self-vs-other effect.

4.3 Diversity: a secondary scaling with the number of training individuals

Consistent with that secondary effect, sweeping the number of training mice $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 1, right) shows pooled aging becoming mildly more favorable as N grows for lick ($-0.001 \rightarrow -0.010$), reward ($-0.004 \rightarrow -0.010$), and tone ($+0.003 \rightarrow -0.010$); for lever-pull it saturates immediately (≈ -0.025 from $N{=}1$). The scaling is modest (~ 0.01 AUC) next to the self-vs-other gap (~ 0.03), matching the count-matched decomposition: most of the drift resistance is already present with a single other individual, and additional individuals refine it.

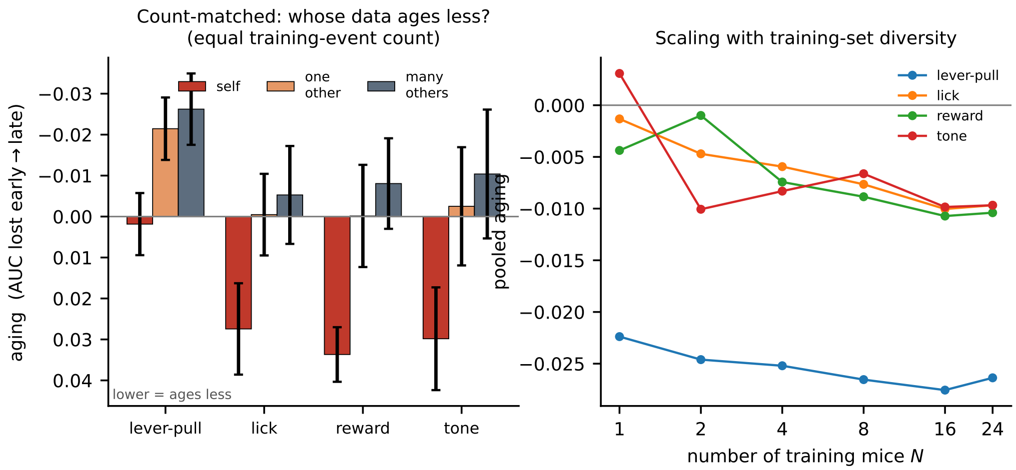

Figure 1: **Left:** count-matched control. At equal training-event count, decoder aging (AUC lost early→late; plotted so lower is more robust) for the subject's own data (self) vs. one other mouse vs. many others. Self ages; other-individual data does not, even at matched count. **Right:** pooled aging vs. number of training mice N; diversity adds a small further gain (saturating for lever-pull).

5 Discussion

On a 25-mouse, 15-day widefield cohort, a decoder pooled over a population of other individuals resists cross-day drift better than a subject's own decoder. A count-matched control shows this is not a data-volume artifact: at equal training-set size, other-individual data ages far less than the subject's own, with cross-individual diversity adding a smaller, consistent refinement. The natural reading is that a subject's own decoder overfits idiosyncratic, drift-prone features of that subject's early sessions, while a decoder that never saw the subject can only latch onto the shared, temporally stable component of the code. This reframes population structure as a *temporal* resource: the well-known small accuracy penalty of cross-subject decoding is not merely tolerable but is repaid, over days, as stability. Where within-subject recalibration spends fresh labelled data from the individual to chase a drifting code, a population prior buys stability up front and for free at test time.

Limitations. The effect is measured at the mesoscale (parcel-averaged) level; whether it holds for single-neuron codes, where drift is characterized, is untested and is the natural next question. We study four thresholdable task events over a two-week window in one species; longer horizons, graded behaviors, and other modalities remain open. Our conclusions rest on within-vs-pooled and early-vs-late *comparisons* rather than absolute AUC. Finally, the count-matched control rules out data volume and points to self-overfitting, but the diversity increment is small and we do not identify which components of the drift are individual-specific.

Reproducibility

Data are open (BraiDyn-BC, DANDI:001425). We release the analysis code and a streaming feature pipeline that reproduces every number from the public parcellated traces without downloading raw imaging; all splits, permutation seeds, and hyperparameters are in Section 3 and the released code.

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